

✓ Human Capital Migration Analysis

Project Done by - Sanskar Shrivastava

Date of completion - xx/05/2026

Data location :-

<https://www.kaggle.com/datasets/daviddave7100/immigrations-and-emigrations-statistics>

✓ Step 1 - Library and data import and initialization

```
#1.1 - Fetching important libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

```
#1.2 - Getting the dataset from the files
path=r'/content/data.csv'
df=pd.read_csv(path)
df.head()
```

	Country	Population	Emigrants Population	Immigrant Population	Net Population Change	Net Population Change to Total Population Ratio
0	Andorra	77700	11132	45574	34442	44.33%
1	Luxembourg	627704	81757	298062	216305	34.46%
2	San Marino	34,007	2407	5543	3136	9.22%

Next steps:

[Generate code with df](#)

[New interactive sheet](#)

✓ Step 1.6 - Data Cleaning: Convert relevant columns to numeric types

```
#1.3 - Fetching number of rows and columns in the dataset  
df.shape
```

```
(201, 10)
```

```
#1.4 - Checking for null values  
df.isnull().sum()
```

	0
Country	0
Population	0
Emigrants Population	0
Immigrant Population	0
Net Population Change	0
Net Population Change to Total Population Ratio	0
Outbound mobility ratio	0
Emmigrant to Population Ratio	0
Immigrants to Population Ratio	0
GDP per capita	0

```
dtype: int64
```

```
#1.5 - Checking for true-duplicates  
df.duplicated().sum()
```

```
np.int64(0)
```

✓ Step 2 - Exploratory Data Analytics

```
#2.1 - Statistical insights for the data  
df.describe()
```

	Emigrants Population	Immigrant Population	Net Population Change
count	2.010000e+02	2.010000e+02	2.010000e+02
mean	1.324783e+06	1.376849e+06	5.206545e+04
std	2.226064e+06	4.177112e+06	4.264051e+06
min	1.740000e+02	2.390000e+02	-1.299079e+07
25%	1.329650e+05	4.954900e+04	-7.817090e+05
50%	6.401600e+05	2.522280e+05	-6.196600e+04
75%	1.489823e+06	1.067090e+06	1.177670e+05
max	1.786949e+07	5.063284e+07	4.763661e+07

```
#2.2 - Which countries have the highest and lowest net population change?
print("Maximum :")
res=df[['Country','Net Population Change']].sort_values(by='Net Populatio
print(res)

print("Minimum :")
res=df[['Country','Net Population Change']].sort_values(by='Net Populatio
print(res)
```

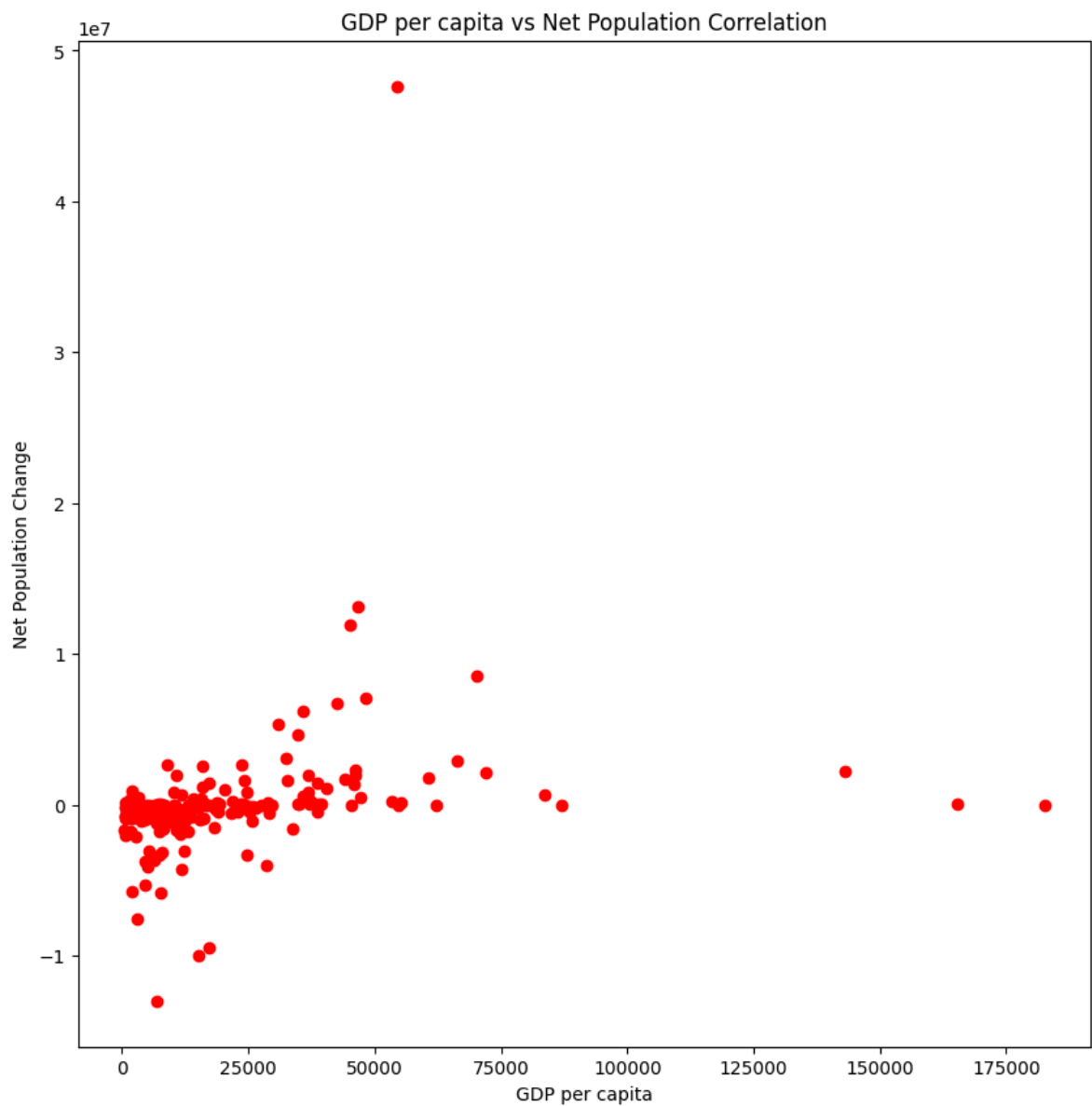
```
Maximum :
Country                United States
Net Population Change    47636613
Name: 111, dtype: object
Minimum :
Country                India
Net Population Change   -12990788
Name: 99, dtype: object
```

```
# Convert GDP per capita and Net Population Change to float
df['GDP per capita']=df['GDP per capita'].astype(str).str.replace(',','')
df['Net Population Change']=df['Net Population Change'].astype(str).str.r
```

```
#2.3 - GDP Per capita vs Net Population Correlation
res = df['GDP per capita'].corr(df['Net Population Change'])
print(res)

x=df['GDP per capita']
y=df['Net Population Change']
plt.figure(figsize=(10,10))
plt.scatter(x,y,color='red')
plt.xlabel('GDP per capita')
plt.ylabel('Net Population Change')
plt.title('GDP per capita vs Net Population Correlation')
plt.show()
```

0.2709921768359454



```
#2.4 - Identify countries with the highest and lowest values for Outbound
res=df.sort_values(by='Outbound mobility ratio',ascending=False).iloc[0][
print("Highest Outbound mobility ratio :\n"+res)
print('-----')
res=df.sort_values(by='Outbound mobility ratio',ascending=True).iloc[0][
print("Lowest Outbound mobility ratio :\n"+res)
print('=====')
```

```

res=df.sort_values(by='Emmigrant to Population Ratio',ascending=False).il
print("Highest Emigration to population ratio :\n"+res)
print('-----')
res=df.sort_values(by='Emmigrant to Population Ratio',ascending=True).ilo
print("Lowest Emigration to population ratio :\n"+res)
print('=====')
res=df.sort_values(by='Immigrants to Population Ratio',ascending=False).i
print("Highest Emigration to population ratio :\n"+res)
print('-----')
res=df.sort_values(by='Immigrants to Population Ratio',ascending=True).il
print("Lowest Emigration to population ratio :\n"+res)

```

```

Highest Outbound mobility ratio :
Turkmenistan - 95.54%
-----
Lowest Outbound mobility ratio :
Dominica - 0.00%
=====
Highest Emigration to population ratio :
Honduras - 9.92%
-----
Lowest Emigration to population ratio :
Papua New Guinea - 0.05%
=====
Highest Emigration to population ratio :
Portugal - 9.83%
-----
Lowest Emigration to population ratio :
Cuba - 0.03%

```

```

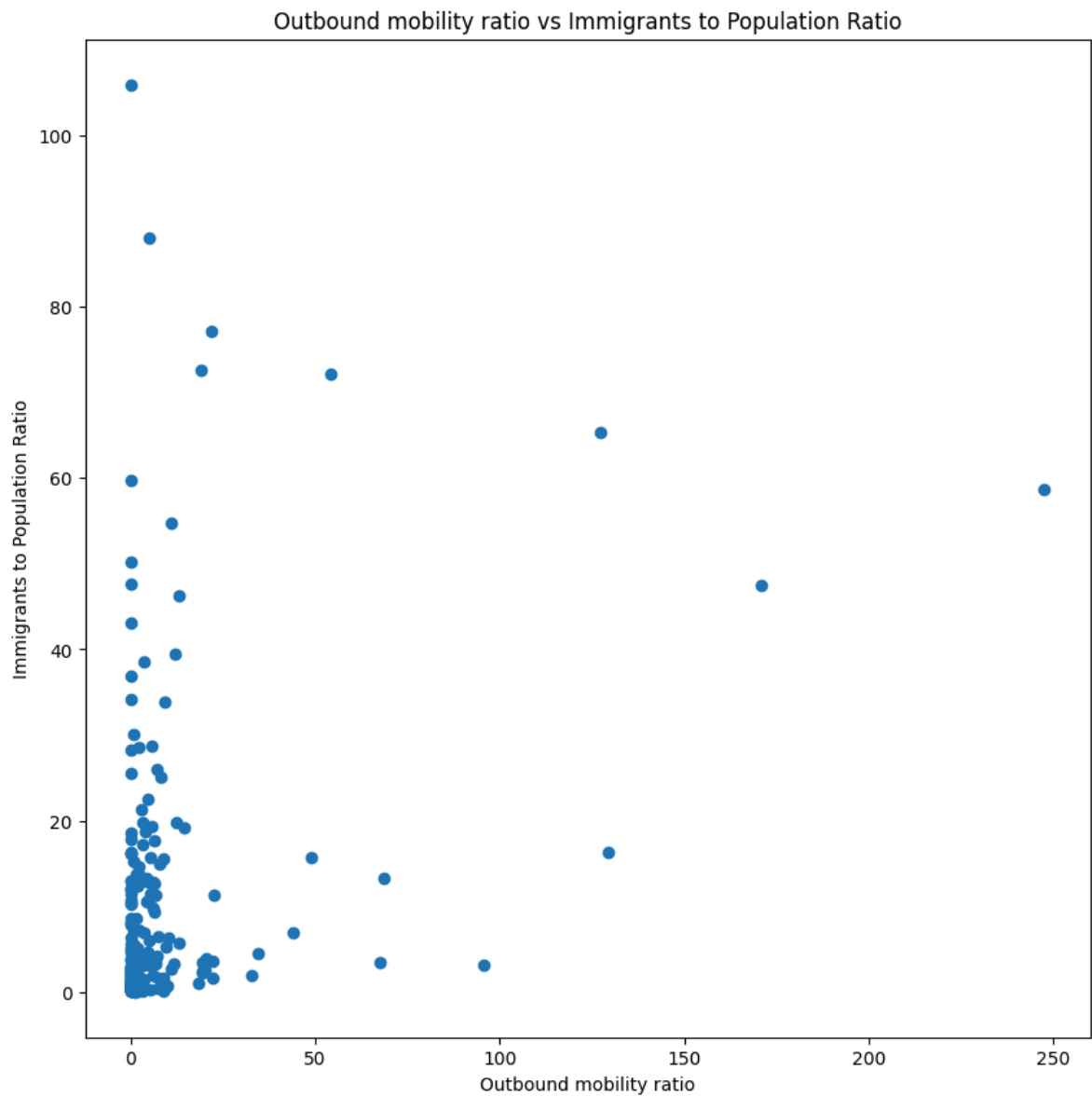
# Convert percentage columns to float
df['Outbound mobility ratio']=df['Outbound mobility ratio'].astype(str).s
df['Immigrants to Population Ratio']=df['Immigrants to Population Ratio']
df['Emmigrant to Population Ratio']=df['Emmigrant to Population Ratio'].a
df['Population']=df['Population'].astype(str).str.replace(',','').astype

```

```

#2.5 - How are the Outbound mobility ratio and Immigrants to Population R
x=df['Outbound mobility ratio']
y=df['Immigrants to Population Ratio']
plt.figure(figsize=(10,10))
plt.scatter(x,y)
plt.xlabel('Outbound mobility ratio')
plt.ylabel('Immigrants to Population Ratio')
plt.title('Outbound mobility ratio vs Immigrants to Population Ratio')
plt.show()

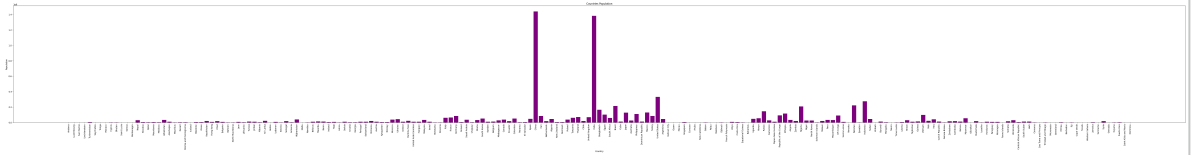
```



✓ Step 3 - Visualization

```
#3.1 - countries population  
x=df['Country']  
y=df['Population']
```

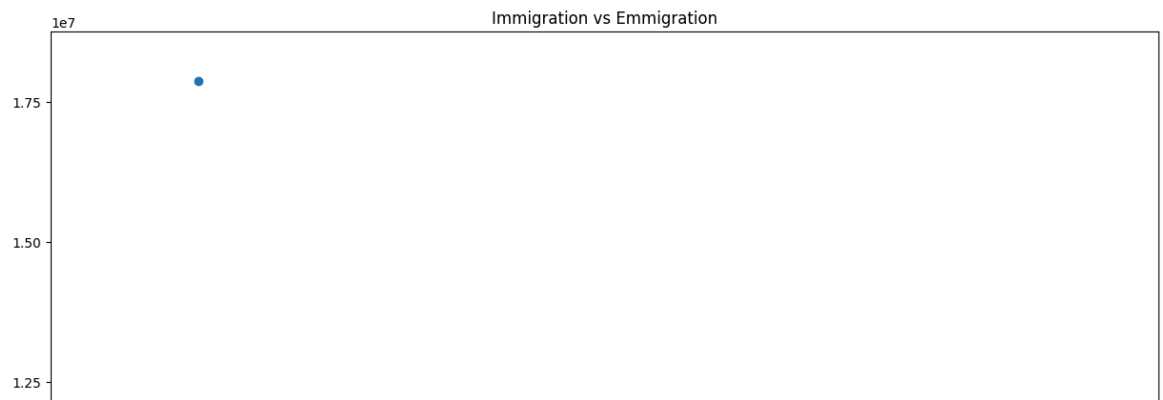
```
plt.figure(figsize=(100,10))
plt.bar(x,y,color='purple')
plt.xlabel('Country')
plt.ylabel('Population')
plt.title('Countries Population')
plt.xticks(rotation=90)
plt.show()
```



```
#3.2 - Immigration vs Emmigration chart
x=df['Immigrant Population']
y=df['Emigrants Population']
plt.figure(figsize=(15,15))
plt.scatter(x,y)
plt.xlabel('Immigrant Population')
plt.ylabel('Emmigrant Population')
plt.title('Immigration vs Emmigration')
```



Text(0.5, 1.0, 'Immigration vs Emmigration')



#3.3 - Net population change vs GDP

```
x=df['Net Population Change']
```

```
y=df['GDP per capita']
```

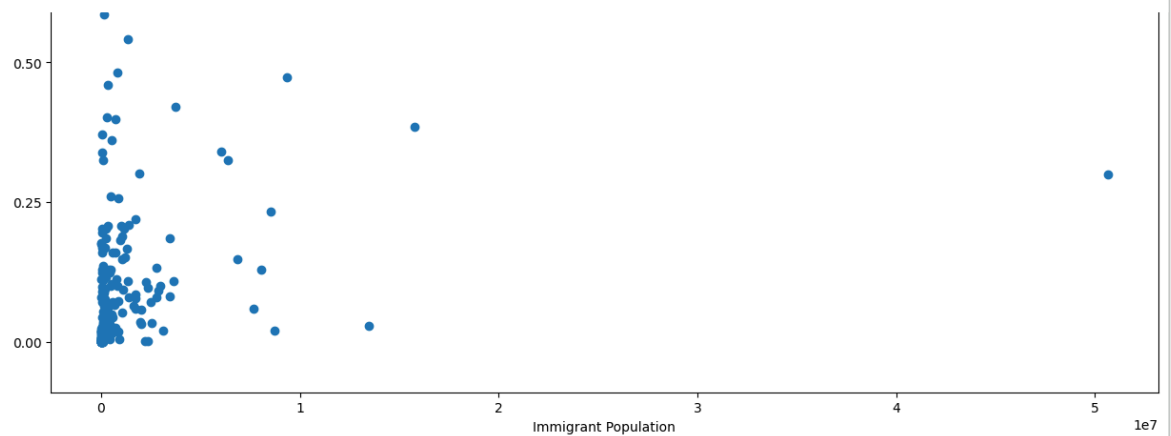
```
plt.figure(figsize=(15,15))
```

```
plt.scatter(x,y)
```

```
plt.xlabel('Net Population Change')
```

```
plt.ylabel('GDP per capita')
```

```
plt.show()
```





```

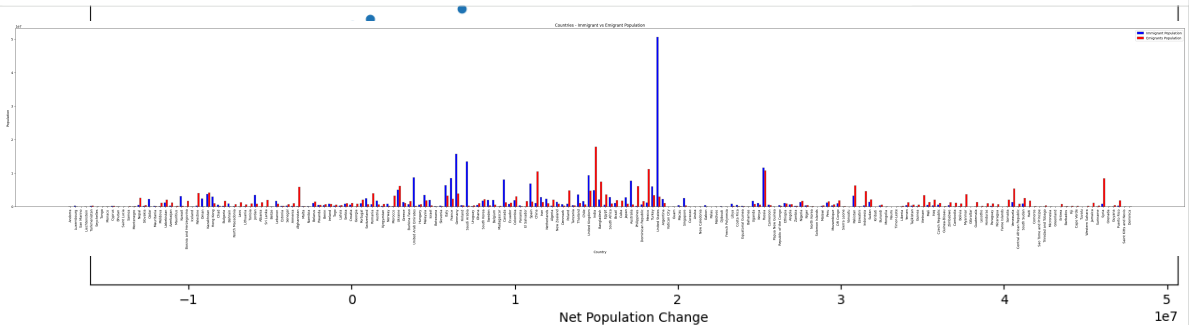
#3.4 - country wise emigration and immigration bar plot
x = df['Country']
y = df['Immigrant Population']
z = df['Emigrants Population']

bar_width = 0.35
r1 = np.arange(len(x))
r2 = [x_val + bar_width for x_val in r1]

plt.figure(figsize=(50, 10))
plt.bar(r1, y, color='blue', width=bar_width, edgecolor='grey', label='Im
plt.bar(r2, z, color='red', width=bar_width, edgecolor='grey', label='Emi

plt.xlabel('Country')
plt.ylabel('Population')
plt.title('Countries - Immigrant vs Emigrant Population')
plt.xticks([r + bar_width / 2 for r in range(len(x))], x, rotation=90)
plt.legend()
plt.tight_layout()
plt.show()

```



```

#boxplot for each field

```